



# **INTERNSHIP PRESENTATION**

**Mayank Kumar**

**Roll No.: 25SCS1003004705**

**Section: CSE 32**

## **INTERNSHIP DETAILS**

**Domain: Artificial Intelligence**

**Organization: Codec Technologies**



## PROOF OF INTERNSHIP

# Certificate & Offer Letter



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### INTERNSHIP OFFER LETTER

Dear Mayank Kumar,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 29/06/2026 to 29/07/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.  
Best regards,

Sincerely,  
Dr. Anurag Shrivastava,  
Codec Technologies India  
NCS ID: E19E86-0116588288923  
Date Of Issue - 29/06/2026

  
**Dr. Anurag Shrivastava**  
Talent Acquisition Manager



 [www.codectechnologies.in](http://www.codectechnologies.in)  [support@codectechnologies.in](mailto:support@codectechnologies.in)  Chandivali IT hub, Mumbai

Offer Letter — Codec Technologies

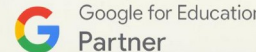


## CERTIFICATE OF INTERNSHIP

This certificate is awarded to  
**Mayank Kumar**

**From Codec Technologies Pvt. Ltd.**

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC  
Approved internship program as  
**Artificial Intelligence Intern**  
Conducted from 29/06/2026 to 29/07/2026

  
**Dr. Anurag Shrivastava**  
Program Manager

AICTE ID - CORPORATE6759d549ce59e1733940553

Certificate of Internship

MACHINE LEARNING INTERNSHIP

# Building a Beginner-Friendly Chatbot

*A project built during my ML Internship at Codec Technologies Pvt. Ltd.*

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**Mayank Kumar**

Machine Learning Intern | 29 Jun 2026 – 29 Jul 2026

# Where I Did This Project

|          |                                       |
|----------|---------------------------------------|
| Company  | Codec Technologies Pvt. Ltd.          |
| Role     | Machine Learning Intern               |
| Duration | 29 June 2026 – 29 July 2026 (1 Month) |
| Mode     | Pan India, Hybrid                     |
| Program  | AICTE & ICAC Approved Internship      |

## WHY THIS INTERNSHIP

It gave me hands-on, practical exposure to how AI/ML projects are actually built — from picking a simple problem to shipping a working demo.

Google for Education Partner

ISO 9001 Certified

National Internship Portal

# Problem Statement

Most beginners find it hard to understand how conversational AI works. The goal was to design and build a simple chatbot that can understand basic user messages and reply automatically — without needing complex, heavy AI models.



## No instant replies

Users often need quick answers to common questions, but no one is always available to respond.



## Learning by building

As a beginner, the best way to understand NLP and chat systems is to build one from scratch.



## Keep it simple

Focus on a lightweight, easy-to-explain chatbot rather than a complex, hard-to-debug system.

## GOAL

# Objective

- 1 Build a simple, rule-based chatbot using Python
- 2 Make it understand common greetings, questions & keywords
- 3 Generate relevant automatic replies in real time
- 4 Learn the basic building blocks of NLP & conversational AI
- 5 Deliver a working, demo-ready beginner project

# Tools & Technologies Used



## Python

Core programming language used to write the chatbot logic.



## Keyword / Rule Matching

Simple if-else and keyword-matching logic to understand user input.



## String Handling (NLP basics)

Lowercasing, cleaning and matching text — the beginner-level NLP used.



## Jupyter / VS Code

Environment used to write, run and test the chatbot step by step.



## Command Line Interface

The chatbot runs and chats with the user directly in the terminal/console.



## Google Colab / Local Setup

Used for quick testing during the internship's hybrid mode.

# Methodology — How I Built It

01

## Collect Sample Inputs

Listed common things users might type — greetings, questions, thank-yous, goodbyes.

02

## Clean the Text

Converted input to lowercase and removed extra spaces/punctuation so matching is reliable.

03

## Match Keywords

Checked the cleaned input against predefined keyword lists using simple if-else conditions.

04

## Pick a Response

For each matched keyword group, the bot picks a suitable reply from a small response set.

05

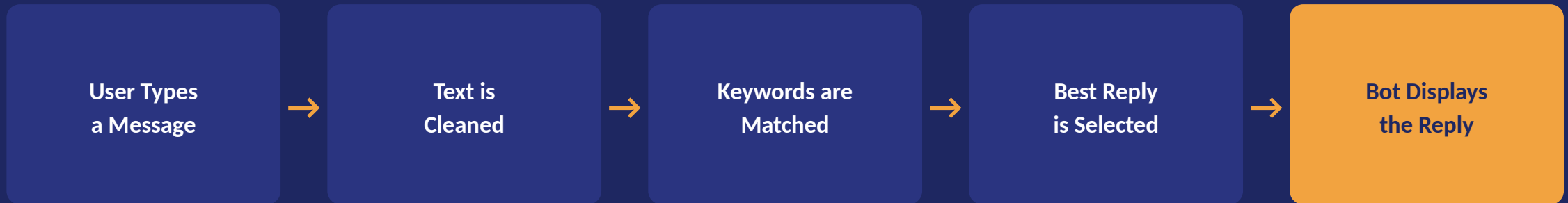
## Chat in a Loop

Wrapped everything in a while loop so the bot keeps chatting until the user types 'bye'.



## HOW IT WORKS

# Chatbot Flow



*Example: if the input contains “hello” or “hi” → the bot replies with a greeting. If it contains “bye” → the chat loop ends.*

# Sample Conversation

## CHATBOT — CONSOLE

Hi there!

Hello! How can I help you today?

What is your name?

I'm a simple chatbot built by Mayank.

Thank you

You're welcome! Happy to help.

Bye

Goodbye! Have a great day.

## Key Features

- Understands greetings & basic questions
- Instant, automated text replies
- Runs continuously until user says 'bye'
- Lightweight — no heavy libraries needed
- Easy to extend with more keywords later

## TAKEAWAYS

# Challenges & Learnings

### CHALLENGES FACED

- Handling different ways users phrase the same question
- Avoiding the bot giving no reply for unmatched input
- Keeping keyword lists organized as they grew

### WHAT I LEARNED

- Basics of how chatbots and NLP systems work
- Writing clean, structured Python logic
- Debugging and testing conversational flows
- Presenting a technical project simply

# Thank You

*Happy to answer any questions about the project or my internship experience.*

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Mayank Kumar | Machine Learning Intern | Codec Technologies Pvt. Ltd.